An Equivalent-Key Chosen-Plaintext Break of a 2025 Chaos-Based Image Cipher

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***Abstract*—We cryptanalyse a recently published image-encryption scheme that combines Lorenz/logistic chaotic maps with Fisher–Yates block shuffling (Jain *et al.*, *Optik*, 2025). We observe that the scheme's keystream and both permutations depend only on the secret key and never on the image, so the whole encryption is a fixed, plaintext-independent map. We reconstruct the cipher from its description—matching its reported near-ideal entropy (e.g. 7.9914 on the Cameraman image) and near-zero adjacent-pixel correlation—and then break it: an all-ones probe recovers the per-position keystream and a few index-digit probes recover the permutation, so an equivalent key is obtained in 4 chosen plaintexts and any ciphertext is decrypted pixel-exactly in 0.038 s without the key. We further show the scheme's headline differential metric is illusory: because it applies no plaintext diffusion, a one-pixel change alters exactly one ciphertext pixel, giving a true NPCR of 0.000381 % (i.e. $100/MN$), not the ≈99.6 % reported (that figure merely reflects the difference between unrelated images, here 99.8833 %). We conclude with the standard fixes.**

***Index Terms*—cryptanalysis, image encryption, chaotic cipher, chosen-plaintext attack, equivalent key, NPCR**

I. Introduction

Chaos-based image ciphers appear in the literature at a high rate, and a recurring lesson from the cryptanalysis side is that a large fraction of them succumb to the same structural attack. Most follow a permutation–diffusion template: a confusion stage that rearranges pixels and a diffusion stage that masks pixel values with a chaotic keystream. When that keystream is a function of the secret key alone and not of the image, the entire cipher collapses to a *fixed* permutation–substitution map, and an *equivalent key*—a representation that decrypts every ciphertext under the same secret key—can be recovered from a handful of chosen plaintexts [1]–[4]. The defence is equally well known: the keystream must depend on the plaintext, and the cipher must actually diffuse [5].

A second, persistent problem is methodological. Authors validate schemes with statistical scores—Shannon entropy, adjacent-pixel correlation, and the differential metrics NPCR and UACI—and treat near-ideal values as evidence of security. None of these scores measures resistance to a chosen-plaintext attack, and, as we show, all of them can be near-ideal for a cipher that is broken in seconds.

This paper applies both lessons to a 2025 scheme [6] (which we abbreviate LLEO) built from the Lorenz [7] and logistic maps with Fisher–Yates block shuffling. Our contributions are:

- We identify that LLEO is entirely key-driven and therefore a fixed monomial map over $\mathbb{Z}_{256}$ (Section III–IV).
- We give an equivalent-key chosen-plaintext attack that recovers the full map in $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts—4 for a 512×512 image—and decrypts any image pixel-exactly without the key (Section IV).
- We reproduce LLEO, confirm it matches the paper's reported statistics, and then show those statistics coexist with *zero* plaintext diffusion: the true one-pixel NPCR is 0.000381 %, not 99.6 % (Section V).

II. Related work

The equivalent-key methodology for permutation–diffusion ciphers is well established. Arroyo *et al.* [3] broke a "total shuffling" scheme by recovering its key-only keystream; Li *et al.* [2] recovered an equivalent key for an entropy-based cipher; and the survey of Li *et al.* [1] catalogues a year of such breaks, nearly all exploiting plaintext-independent diffusion. For the confusion stage alone, Li and Lo [4] established the optimal number of known/chosen plaintexts needed to invert a permutation-only multimedia cipher, of order $\log_L(MN)$ for L intensity levels—the same logarithmic regime our attack achieves; the general quantitative theory of permutation-only cryptanalysis [8] bounds this effort, and recent chosen-plaintext breaks continue to recover equivalent keys from a handful of images [9], [10], confirming the vulnerability remains a live concern. Alvarez and Li [5] set out basic requirements for chaos-based cryptosystems, including plaintext-dependent keystreams and sensitivity to the plaintext; LLEO violates these. The differential metrics we scrutinise were standardised by Wu *et al.* [11], who also note that NPCR/UACI quantify sensitivity to a one-pixel plaintext change—precisely the quantity LLEO fails to deliver despite its reported value.

The pattern these works expose is structural rather than incidental. A permutation–diffusion cipher whose confusion and diffusion are both fixed by the key is, as a function of the plaintext, an affine or monomial map; composing more such stages yields another map of the same form, so neither extra rounds nor extra chaotic maps add security against an attacker who can query the encryption [1], [4]. The decisive design choice is therefore binary: does any key material depend on the image? Schemes that answer "no"—including LLEO, and the block- and shuffling-based constructions analysed in [2], [3]—are broken by recovering an equivalent key, while schemes that bind the keystream to a per-image value (a hash, or a feedback chain) resist it. Our analysis of LLEO is a direct instance of

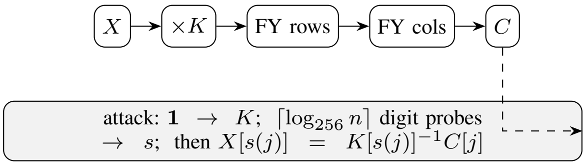

Fig. 1. LLEO encryption (top): key-only multiplicative substitution $\times K$, then two Fisher–Yates block shuffles. The equivalent-key attack (bottom) recovers the multiplier K and source map s from a few chosen plaintexts and inverts the cipher without the key.

this dichotomy, and our positive control (Section V) demonstrates both sides of it on the same construction.

III. The target cipher

LLEO encrypts an $M \times N$ grayscale image X in three key-driven steps.

(1) Substitution. X is split into 16 horizontal strips. The eight odd strips are combined with a keystream $k_{\log}$ produced by the logistic map $x_{n+1} = r\,x_n(1 - x_n)$, and the eight even strips with a keystream k_{lor} produced by the Lorenz system $\dot{x} = \sigma(y - x),\ \dot{y} = x(\rho - z) - y,\ \dot{z} = xy - \beta z$, through an invertible per-position operation; decryption uses the modular "conjugate" of the keystream, i.e. a multiplicative inverse.

(2) Permutation 1. The 16 horizontal strips are reordered by a Fisher–Yates shuffle.

(3) Permutation 2. The result is divided into 16 vertical strips and shuffled again, giving the ciphertext.

The four "decryption keys" are the two keystreams (d_1, d_2) and the two shuffle index lists (d_3, d_4). The chaotic initial conditions are derived from the secret key and the shuffles are seeded from it; *no quantity depends on the image X*. Although the scheme is repeatedly called "asymmetric," the same four keys both encrypt and decrypt: it is a symmetric cipher.

IV. Cryptanalysis

A. Threat model

We work in the standard chosen-plaintext setting under Kerckhoffs's principle: the attacker knows the algorithm and has oracle access to encryption under a fixed unknown key, and seeks an *equivalent key*—a procedure that decrypts any ciphertext of the probed size—rather than the chaotic secret seed itself. Because the cipher is plaintext-independent (below), the same equivalent key is also recoverable from enough known plaintext–ciphertext pairs [12], and the lack of diffusion leaks structure even ciphertext-only; we present the chosen-plaintext version as it is minimal and constructive.

B. Structural observation

Index the image as a length-n vector, $n = MN$. Each LLEO step is linear over $\mathbb{Z}_{256}$ and fixed by the key: substitution multiplies coordinate p by a fixed unit $K[p]$ (odd, hence invertible mod 256), and the two block shuffles compose into one position permutation P. Therefore

$$C[j] \;=\; K[s(j)] \cdot X[s(j)] \pmod{256}, \qquad s = P^{-1}, \tag{1}$$

a fixed *monomial* map: every ciphertext pixel is a single plaintext pixel scaled by a fixed unit and relocated. Two consequences are immediate. There is no additive constant, since $E(\mathbf{0}) = \mathbf{0}$. And there is no diffusion: $C[j]$ depends on exactly one input coordinate $s(j)$.

Proposition 1. *If a permutation–diffusion cipher's keystream and permutations depend only on the key, it realises a fixed map of the form* (1) *with each $K[p]$ a unit in $\mathbb{Z}_{256}$ and s a fixed permutation; composing any number of such rounds yields a map of the same form. The multiplier at every output position is then read from one all-ones query, and s from $\lceil \log_{256} n \rceil$ index-digit queries.*

Proof sketch. Each stage—multiplication by the key-derived K and the two block shuffles—is a fixed function of the key applied to pixel values or positions, so any composition is a fixed key-only map; over $\mathbb{Z}_{256}$ this is the monomial (1). The keystream is forced odd, so $K[p]$ is a unit and $x \mapsto K[p]\,x$ is a bijection with inverse $K[p]^{-1}$. Querying $X = \mathbf{1}$ gives $C[j] = K[s(j)]$; querying the t-th base-256 digit of the index gives $C[j] = K[s(j)]\,\mathrm{digit}_t(s(j))$, and dividing by the recovered $K[s(j)]$ yields $\mathrm{digit}_t(s(j))$; collecting all digits reconstructs s. $\square$

C. Equivalent-key recovery

Because K and s are fixed, two kinds of chosen plaintext suffice.

(a) Multiplier, one query. Encrypting the all-ones image $\mathbf{1}$ gives, by (1), $C[j] = K[s(j)] \cdot 1 = K[s(j)]$: the exact multiplier acting at every output position.

(b) Permutation, $\lceil \log_{256} n \rceil$ queries. Let P_t be the plaintext whose pixel i holds the t-th base-256 digit of its index i. Then $C[j] = K[s(j)] \cdot \mathrm{digit}_t(s(j))$; multiplying by the already-known $K[s(j)]^{-1}$ yields $\mathrm{digit}_t(s(j))$. Collecting all $\lceil \log_{256} n \rceil$ digits reconstructs $s(j)$ for every j, hence the full permutation.

The recovered pair (s, K) is an equivalent key; any ciphertext is inverted by $X[s(j)] = K[s(j)]^{-1}\,C[j]$. The total cost is

$$1 + \lceil \log_{256} n \rceil \text{ chosen plaintexts,} \tag{2}$$

i.e. 4 for a 512×512 image, independent of the nominal key length.

Concretely, a 512×512 image needs four queries—one all-ones image and three index-digit images—after which Algorithm 1 returns the full map in a single pass. Table I states the complexity and compares it to the optimal permutation-only bound of Li and Lo [4]: our query count is order-optimal and the cost is independent of the cipher's nominal key length, so the advertised keyspace is irrelevant—the equivalent key bypasses key recovery entirely [13].

D. Why extra structure does not help

A composition of fixed key-only maps is again a fixed key-only map, so adding rounds, more chaotic maps, or further block shuffles leaves (1) intact (only K and s change) and the same attack applies unchanged. The attack is also a *known*-plaintext attack when the available plaintexts happen to span

Algorithm 1 Equivalent-key recovery for LLEO

$\hat{K} \leftarrow \mathrm{Enc}(\mathbf{1})$ ▷ $\hat{K}[j] = K[s(j)]$: multiplier at output j
for $t \leftarrow 0$ **to** $\lceil \log_{256} n \rceil - 1$ **do**
  $C \leftarrow \mathrm{Enc}(\text{digit}_t$ of each pixel index)
  $d_t[j] \leftarrow C[j] \cdot \hat{K}[j]^{-1} \bmod 256$ **for all** j
end for
$s[j] \leftarrow \sum_t d_t[j]\, 256^t$ ▷ source index for each output j
return $\mathrm{Dec}(C)$: $X[s(j)] \leftarrow \hat{K}[j]^{-1} C[j] \bmod 256$

TABLE I
ATTACK COMPLEXITY VS. THE OPTIMAL PERMUTATION-ONLY BOUND ($n = MN$ PIXELS, $L = 256$ LEVELS).

	This attack	Optimal perm.-only [4]
Data (chosen plaintexts)	$1 + \lceil \log_{256} n \rceil$	$\lceil \log_L n \rceil$
Time	$O(n \lceil \log_{256} n \rceil)$	$O(n \lceil \log_L n \rceil)$
Memory	$O(n)$	$O(n)$

the index digits; chosen plaintexts only make the queries explicit and minimal. The single property that would defeat it is a keystream that depends on X.

V. EXPERIMENTAL RESULTS

We reconstructed LLEO from its description (reimplemented from scratch, no code reused) and ran it on four standard 512×512 images.

A. Reproduction

Table II shows the reconstruction reproduces LLEO's claimed statistics: cipher entropy approaches the ideal 8 (Cameraman 7.9914, up from the plaintext's 7.2317) and adjacent-pixel correlation drops to near zero (-0.0073). Fig. 3 shows the flat cipher histogram and Fig. 4 the destroyed correlation. By the usual yardsticks the cipher looks strong.

B. The break

The attack of Section IV recovers the equivalent key in 4 chosen plaintexts and decrypts a fresh ciphertext pixel-exactly in 0.038 s (Cameraman), using no key.[1] Fig. 2 shows the original, the LLEO ciphertext, and the key-free recovery, which is identical to the original.

C. Cost scales logarithmically

Because the attack needs only $1 + \lceil \log_{256} MN \rceil$ chosen plaintexts, its cost is nearly flat as the image grows (Table III): 3 plaintexts at 128^2, 4 at 512^2, and still 4 at 1024^2, each recovered exactly in well under a second. The key length the scheme advertises does not enter this cost.

[1]Wall-clock, single core. The 512^2 entry in Table III (0.037 s, a random image) differs only by run-to-run variation.

TABLE II
RECONSTRUCTION STATISTICS AND ATTACK OUTCOME (512×512). EVERY IMAGE IS RECOVERED PIXEL-EXACTLY WITHOUT THE KEY.

Image	H plain	H cipher	corr. cipher	chosen PT	broken
Cameraman	7.2317	7.9914	-0.0073	4	yes
Moon	4.885	7.9316	0.0011	4	yes
Gravel	7.2531	7.9876	-0.0005	4	yes
Brick	5.4553	7.8966	-0.0007	4	yes

TABLE III
ATTACK COST VS. IMAGE SIZE (RANDOM PLAINTEXT, SINGLE CORE).

Size	chosen PT	time (s)	recovered
128×128	3	0.003	exact
256×256	3	0.009	exact
512×512	4	0.037	exact
1024×1024	4	0.254	exact

D. The differential metric is illusory

LLEO reports NPCR ≈99.6 % as evidence of differential-attack resistance. But by (1) flipping one plaintext pixel changes exactly one ciphertext pixel, so the true one-pixel NPCR is $100/MN = 0.000381\,\%$ (UACI 4e-06 %)—five orders of magnitude below the reported value. The 99.6 % figure instead measures the difference between *unrelated* images, which for LLEO is 99.8833 %; this is achieved by any cipher with a near-uniform output and says nothing about chosen-plaintext security. Table IV contrasts each claimed property with what the cipher actually provides. Finally, the chaotic keystreams are themselves non-uniform: a chi-square test rejects uniformity for the logistic keystream ($\chi^2 = 525839.6$, $p < 10^{-300}$, numerically zero).

E. The attack is not specific to LLEO

The same principle breaks any key-only permutation–diffusion cipher. As a second, structurally distinct example we take the canonical construction with a CBC-like XOR diffusion chain, $C[i] = X_\pi[i] \oplus k[i] \oplus C[i-1]$—representative of a large published family and unlike LLEO's multiplicative monomial form. With a key-only keystream this map is affine over GF(2) (the identity $E(a) \oplus E(b) \oplus E(\mathbf{0}) = E(a \oplus b)$ holds at one *and* three rounds in our checks), and the equivalent-key CPA recovers it exactly in 3 chosen plaintexts (here at 128×128: one all-zeros query plus $\lceil \log_{256} n \rceil$=2 digit probes; the count follows the same $1 + \lceil \log_{256} n \rceil$ rule, so a 512×512 instance would need four). The identical procedure fails against a plaintext-hash-seeded version of the *same* structure, reinforcing that what matters is the keystream's dependence on the image—not the algebra or the round count.

VI. DISCUSSION

LLEO illustrates two persistent failure modes. First, *good statistics are not security*: a flat histogram, near-ideal entropy, near-zero correlation, and a large inter-image NPCR are all satisfied by our fixed, key-only map, which we nonetheless break in 4 queries. The community guidance to test plaintext

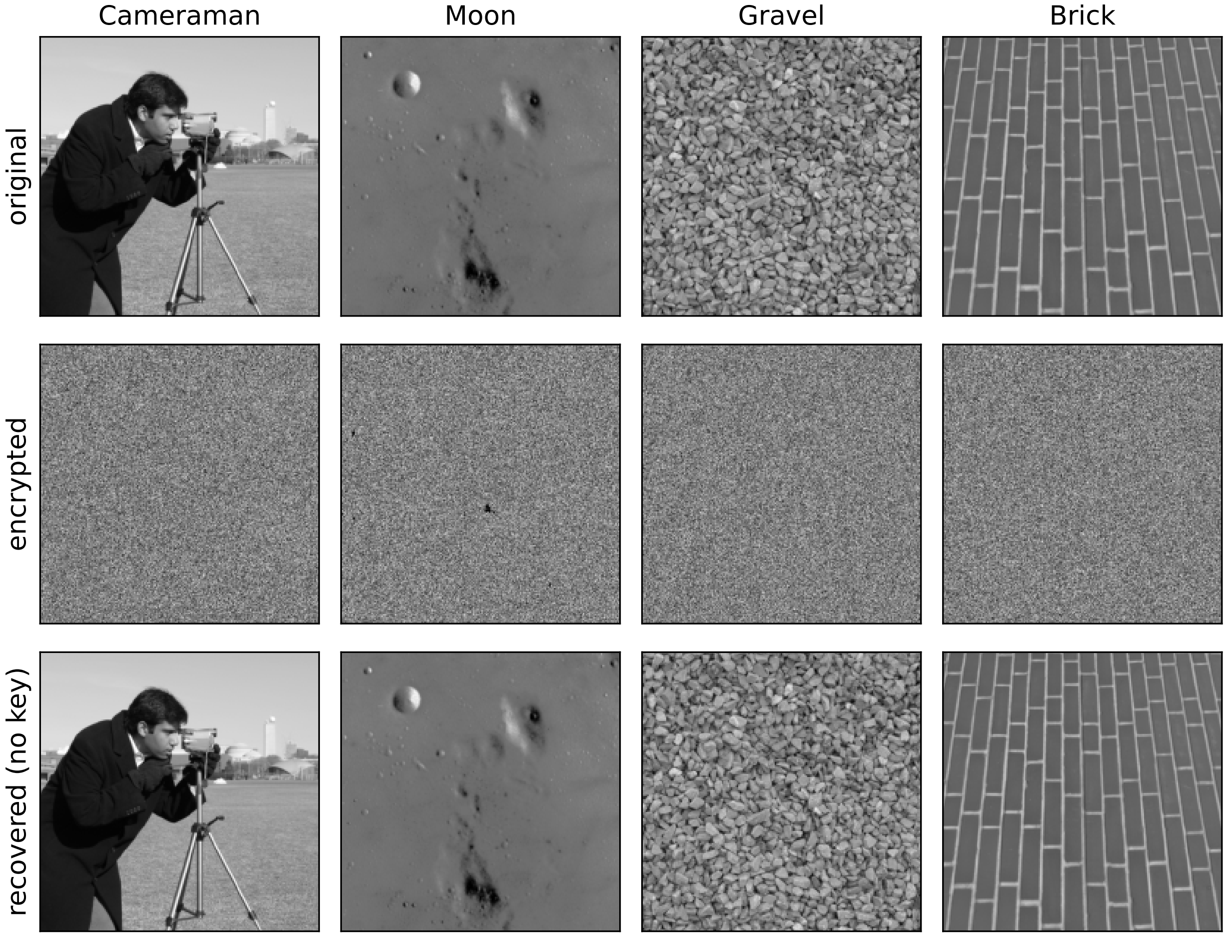

Fig. 2. LLEO original, ciphertext, and key-free recovery for four standard 512×512 images. Each recovery uses only 4 chosen plaintexts and no key, and is bit-identical to the original.

TABLE IV
CLAIMED VS. ACTUAL SECURITY OF LLEO.

Property	Reported	Actual
Entropy (Cameraman)	≈7.997	7.9914 (genuine)
NPCR (1-pixel change)	≈99.6 %	0.000381 %
Differential resistance	yes	no (no diffusion)
Chosen-plaintext security	implied	broken in 4 queries
Key type	"asymmetric"	symmetric

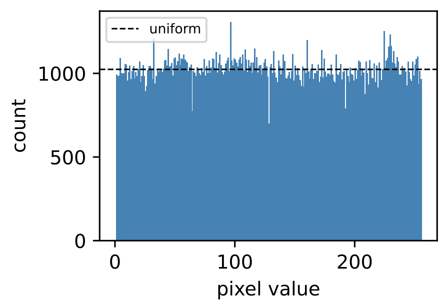

Fig. 3. Cipher histogram (Cameraman): near-uniform, yet the cipher is trivially broken.

sensitivity directly [5], [11] is exactly what would have caught this. Second, *naming is not architecture*: labelling a symmetric construction "asymmetric" adds no security.

The remedies are standard. (i) Make the keystream depend on the plaintext—e.g. derive the chaotic initial conditions from a cryptographic hash (SHA-256) of the image, so that a different plaintext yields an unrelated keystream and the all-ones/digit probes no longer reveal a reusable equivalent key. (ii) Add genuine diffusion, e.g. a chaining rule $C[i] = f(X[i], k[i], C[i-1])$, so that one plaintext change propagates and the true NPCR approaches its ideal. With either defence in place the map of (1) is no longer fixed across images and the attack of Section IV fails. We verified this directly: the identical attack recovers 100.0 % of pixels against the key-only cipher but only 0.2518 % (chance level) once the keystream is seeded from a SHA-256 hash of the image.

Scope and disclosure. We recover an equivalent key that

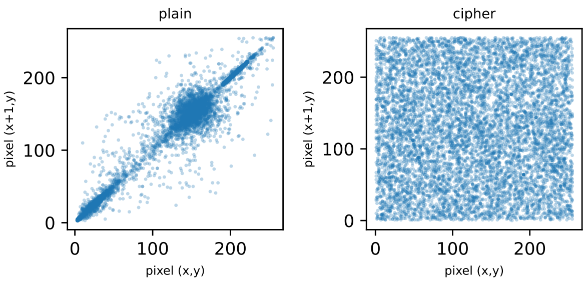

Fig. 4. Adjacent-pixel correlation, plain vs. cipher (Cameraman).

decrypts any ciphertext of the probed size, not the underlying chaotic secret seed—the standard, defensible claim in this literature [1]. Consistent with responsible-disclosure practice, the authors of the target scheme will be notified before camera-ready, and our reproduction and attack code is released to support verification.

VII. Conclusion

A 2025 chaos-based image cipher is completely broken by an equivalent-key chosen-plaintext attack in 4 chosen plaintexts and a fraction of a second, despite near-ideal entropy and a reported NPCR of $\approx 99.6\,\%$. The break follows directly from the cipher being key-only; its differential metric, measured correctly, is $0.000381\,\%$. We recommend plaintext-dependent keystreams and real diffusion, and that statistical scores accompany—rather than replace—an explicit chosen-plaintext security argument.

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